MEMORIA CAM

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3º GRADO EN INGENIERÍA AEROESPACIAL US

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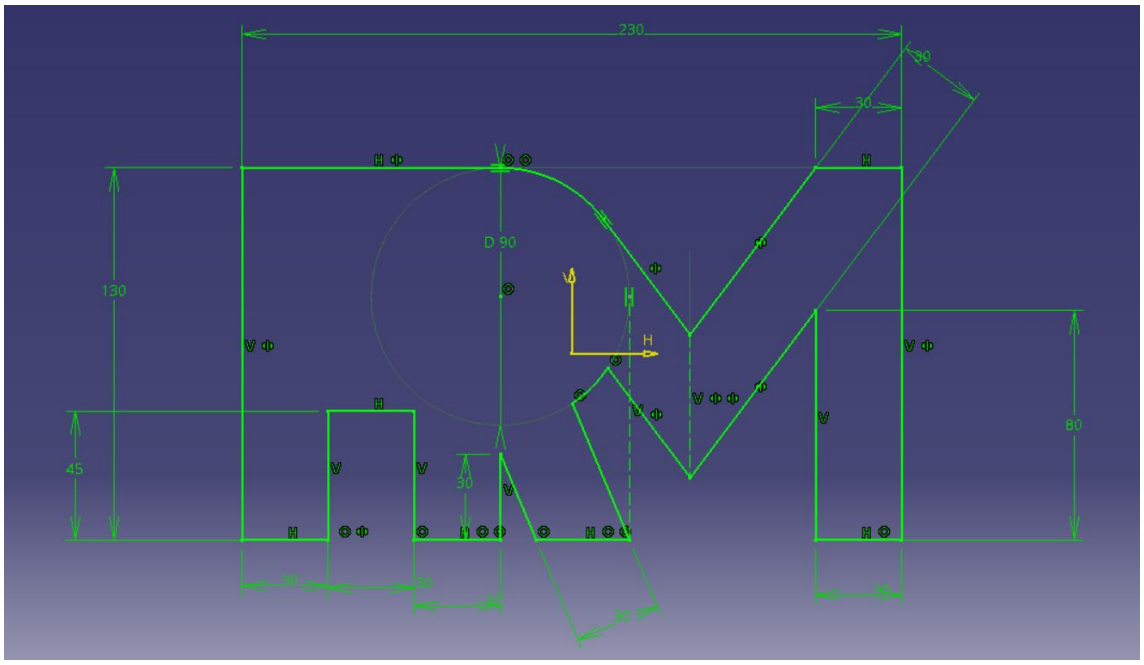
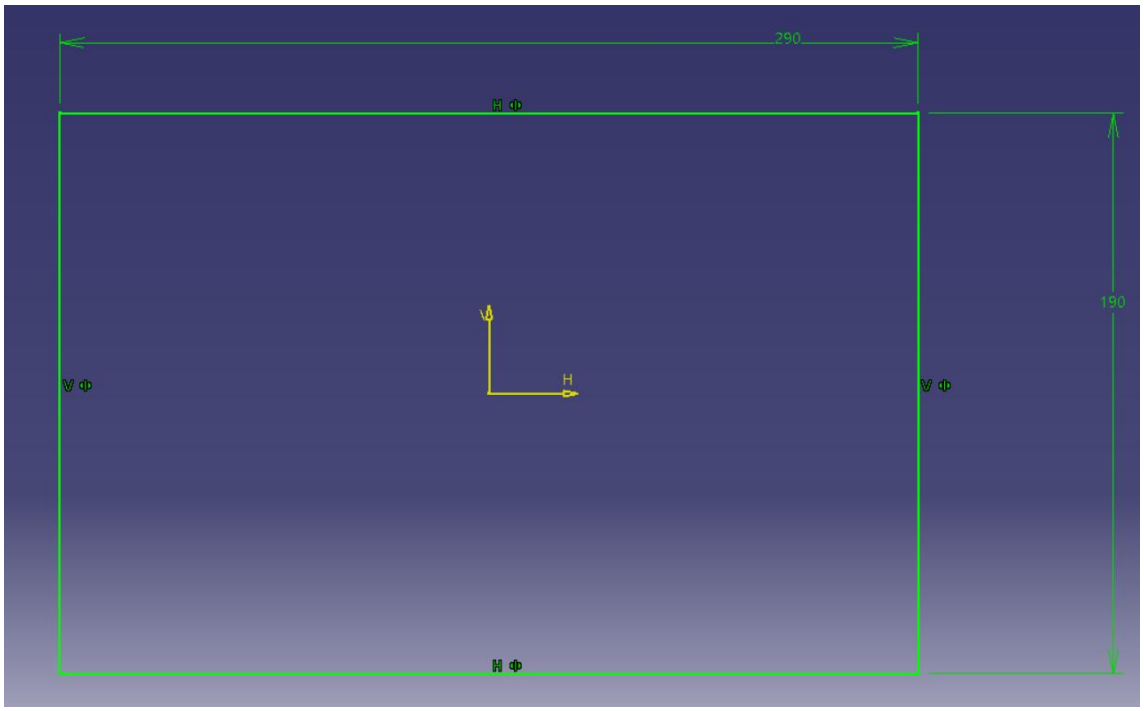
1. Boceto

Boceto de las letras (ARM).



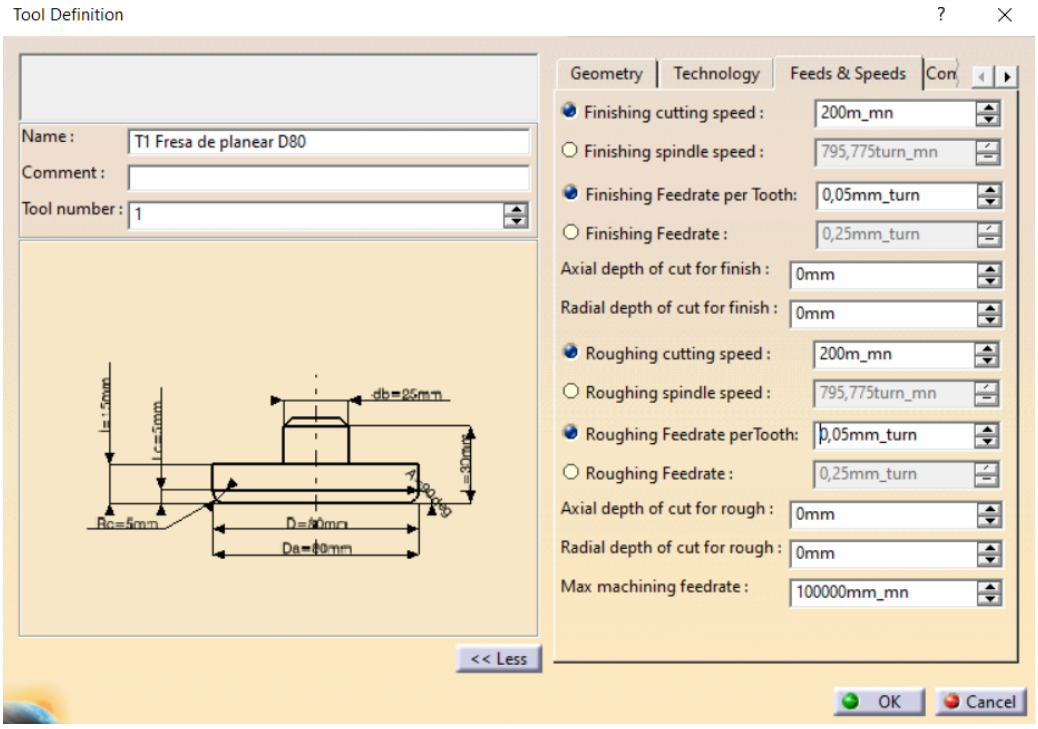
Boceto de las dimensiones de la pieza completa.



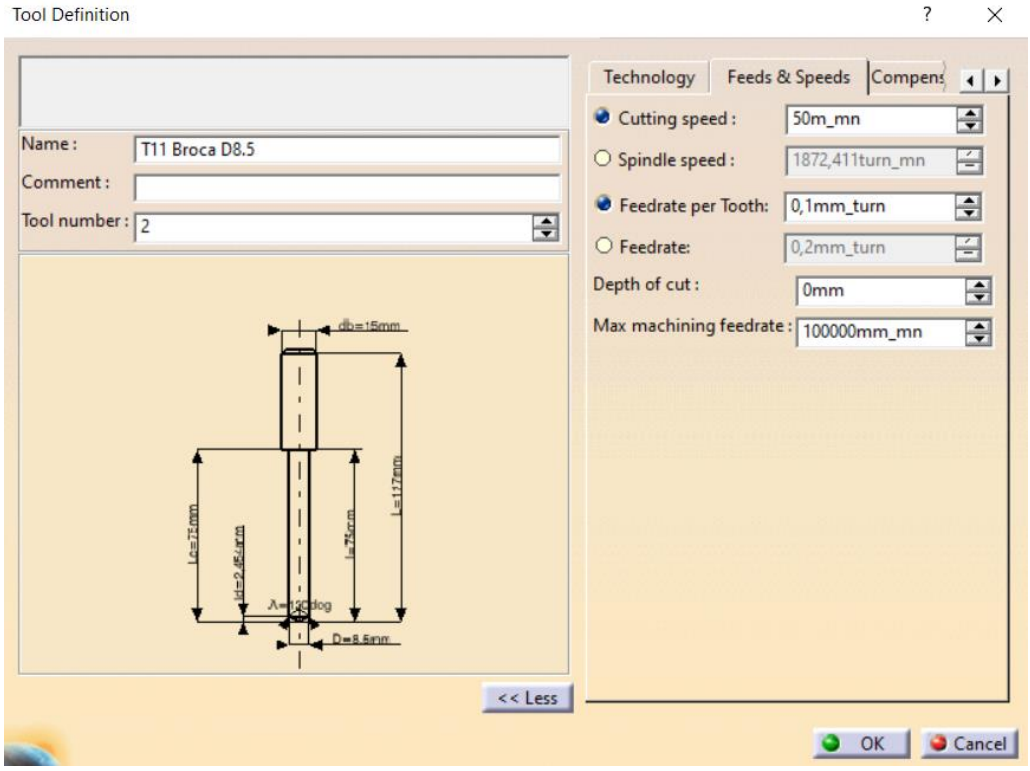
2. Herramientas

Debido a las restricciones de la máquina de trabajo, hemos tenido que reducir las velocidades recomendadas por el fabricante para que no superase la restricción de 4500rpm.

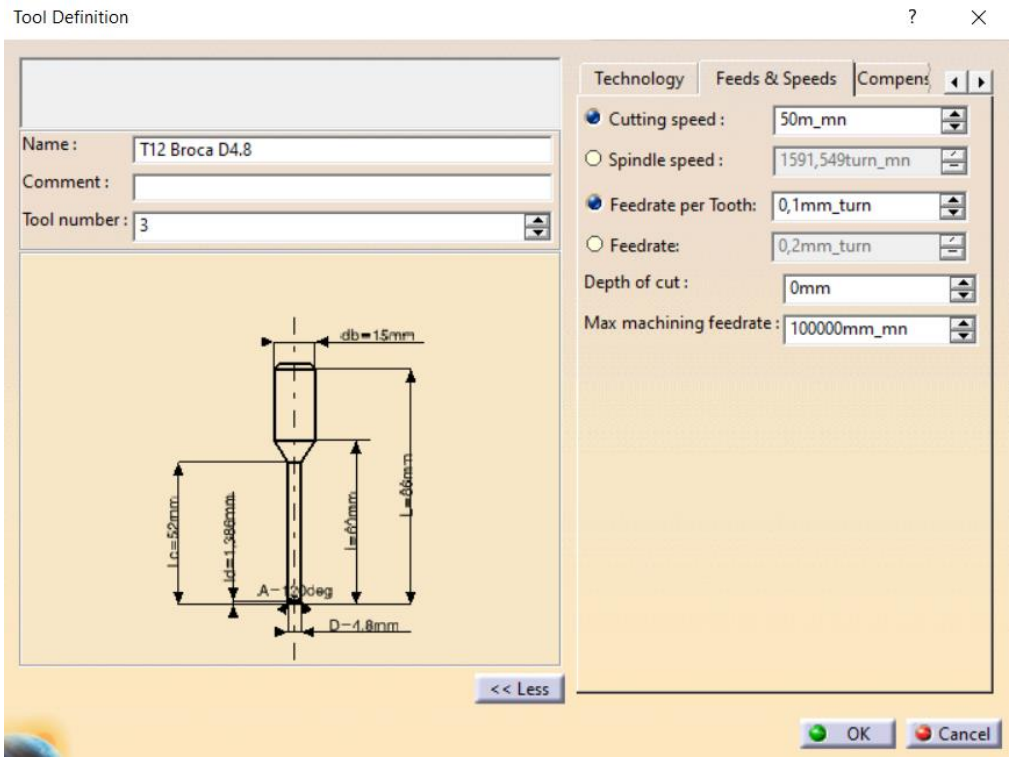
Fresa de planear D80



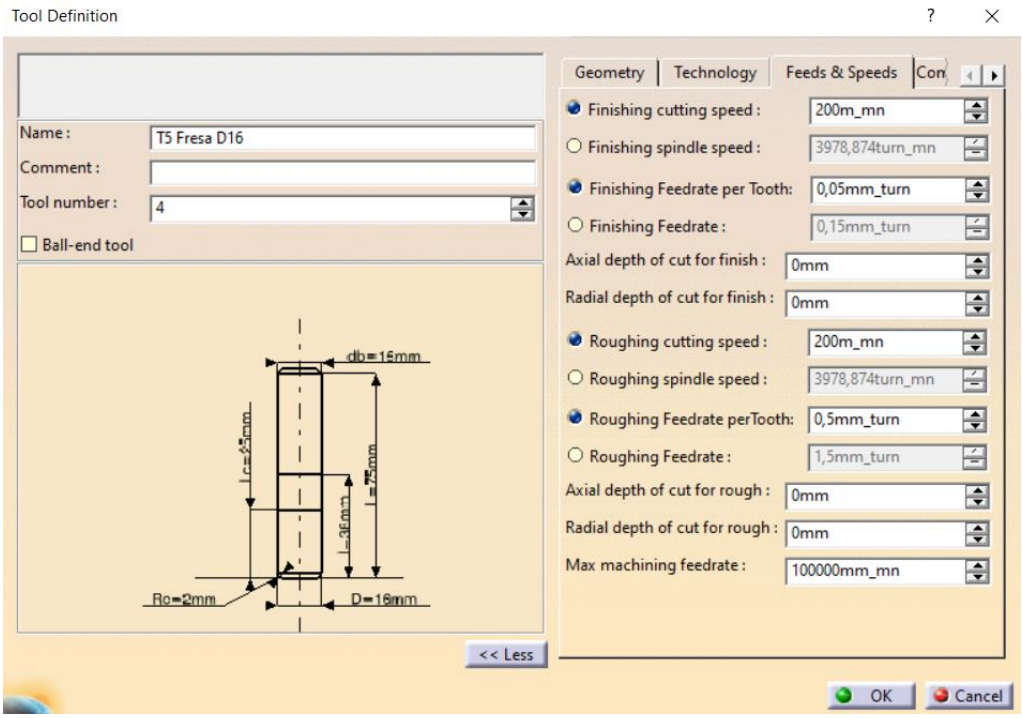
Broca D8.5



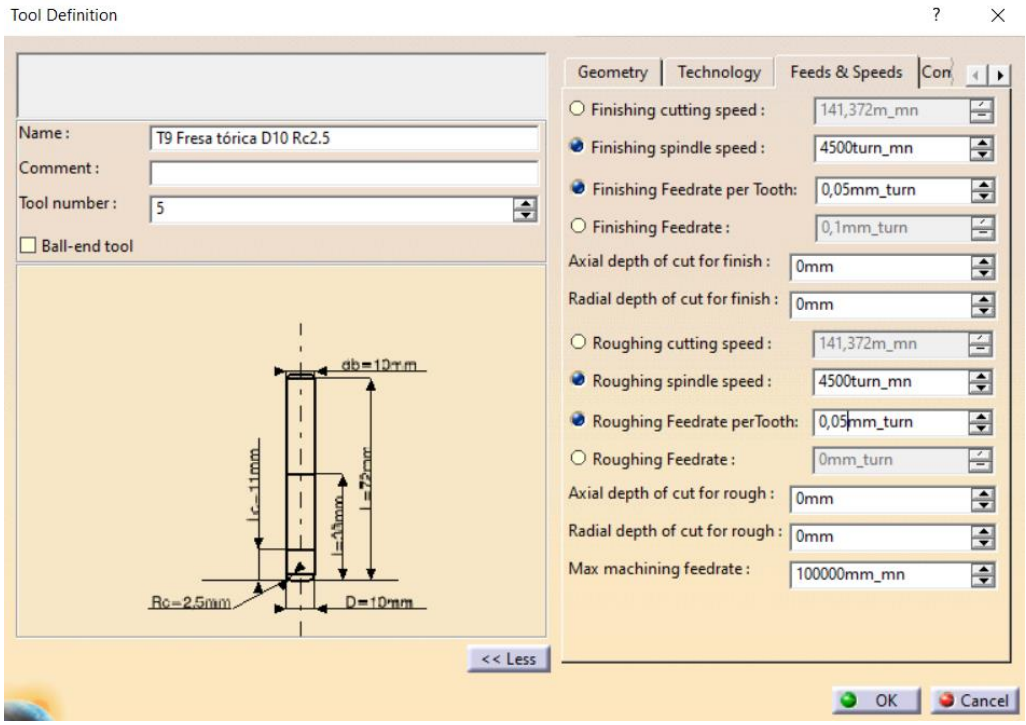
Broca D4.8



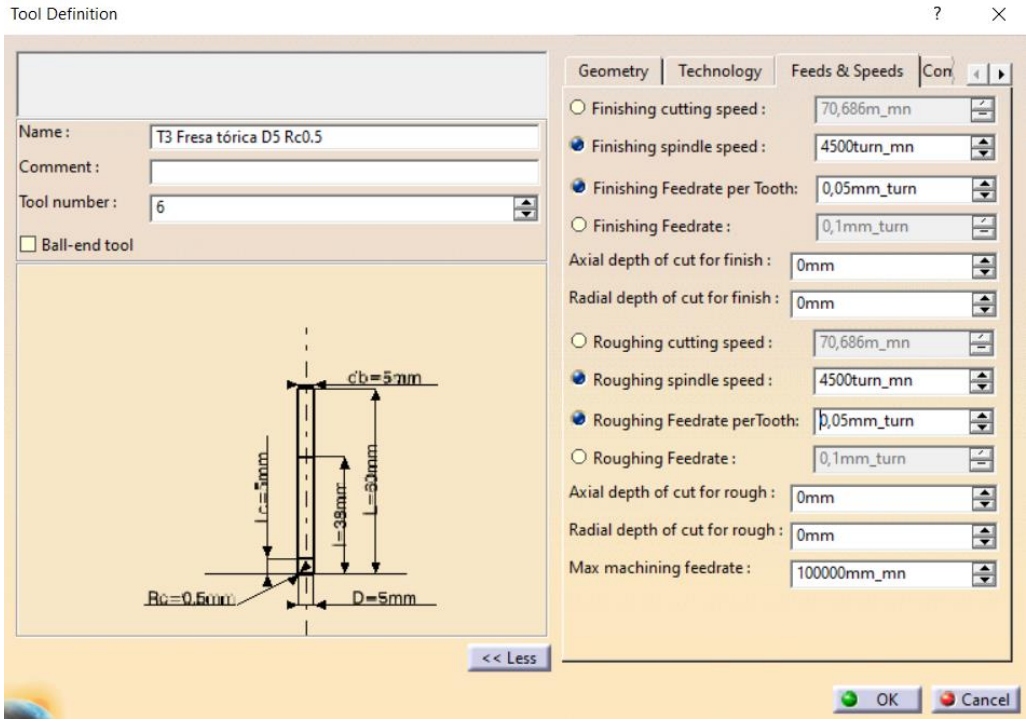
Fresa D16



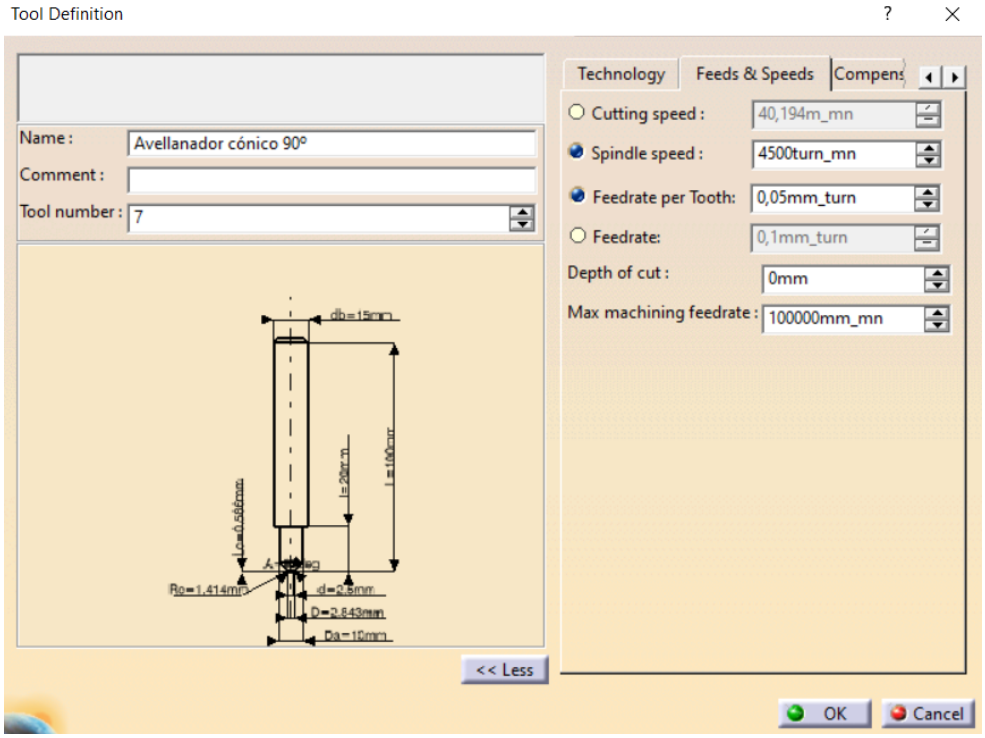
Fresa tórica D10 Rc2.5



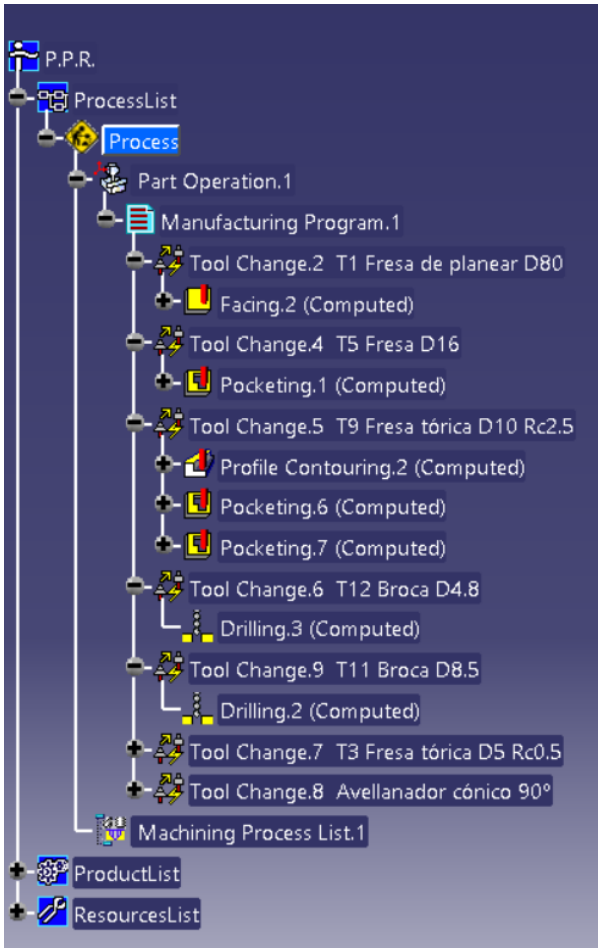
Fresa tórica D5 Rc0.5

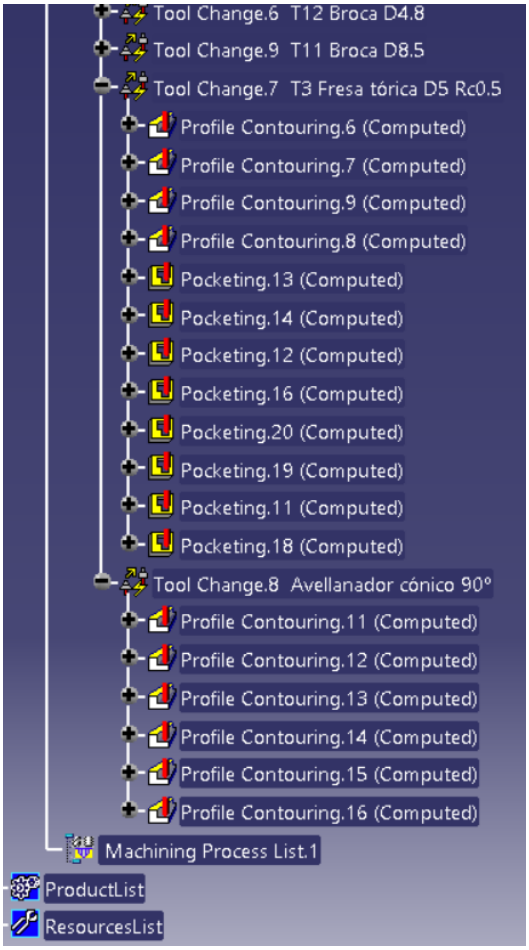


Avellanador cónico 90º



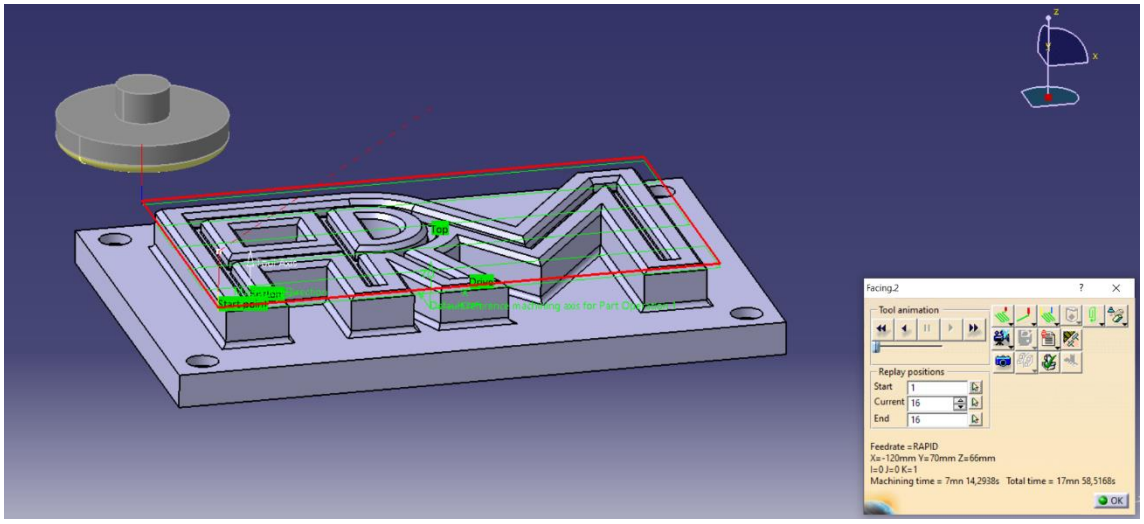
3. Árbol de proceso



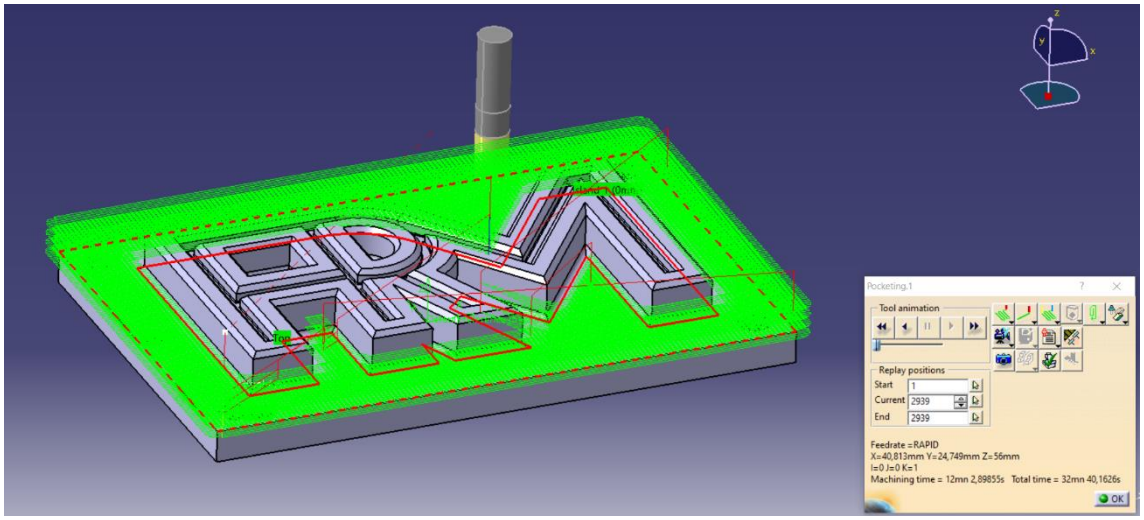


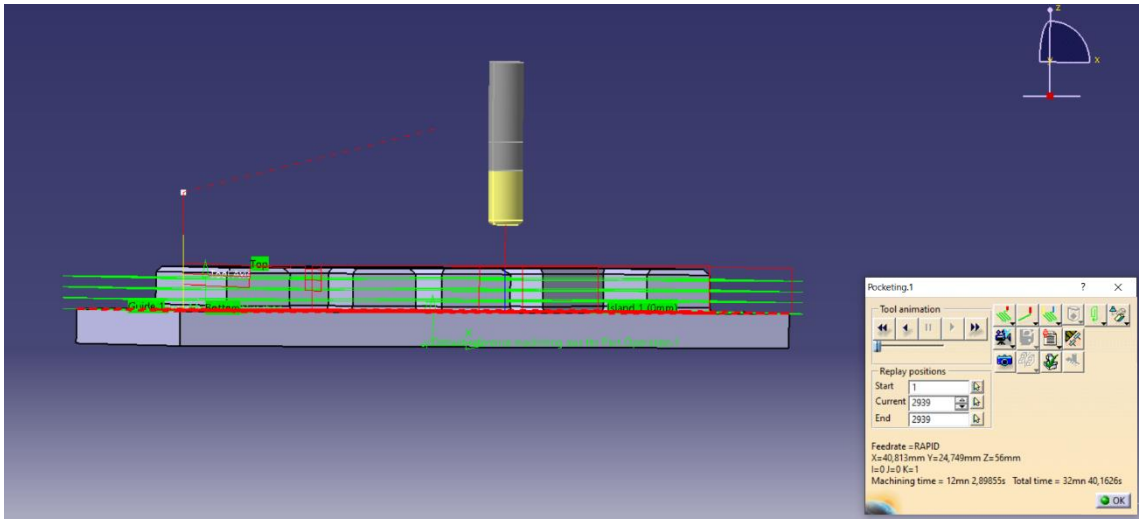
4. Proceso de mecanizado

Planeado con fresa de planear D80

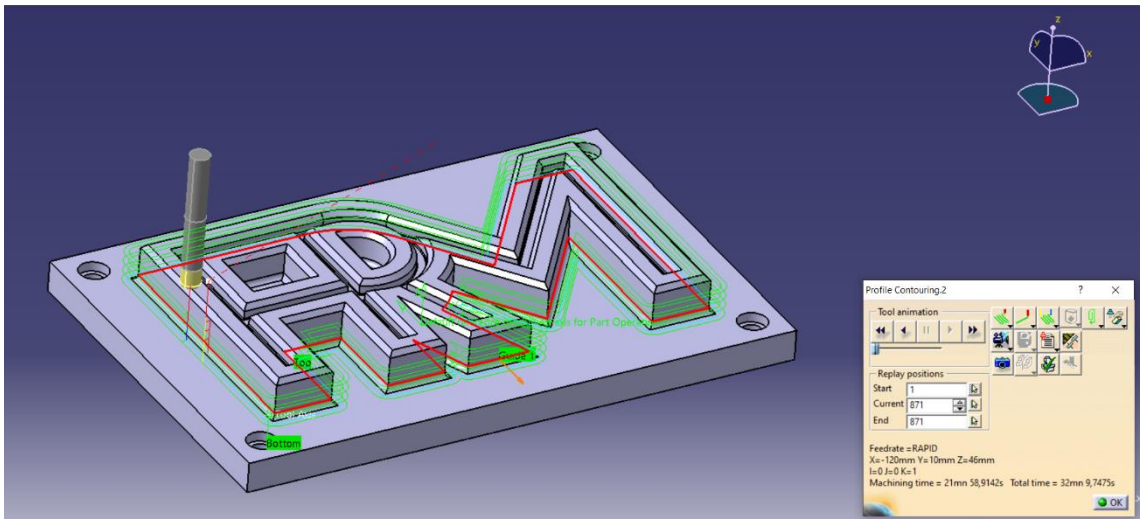


Desbaste con fresa D16 con un porcentaje de sobresalir y de corte de la fresa del 90% para una mayor precisión.

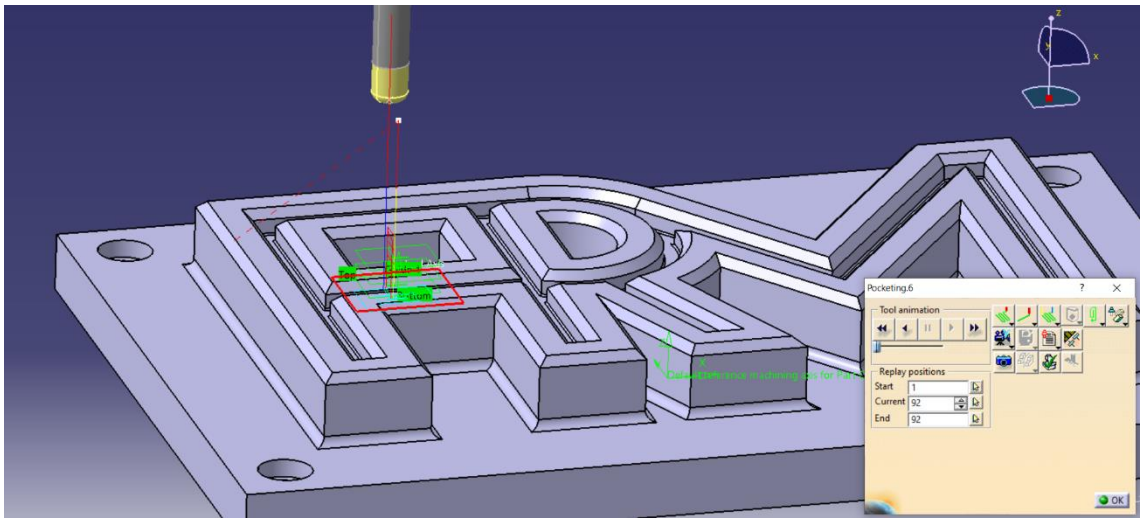


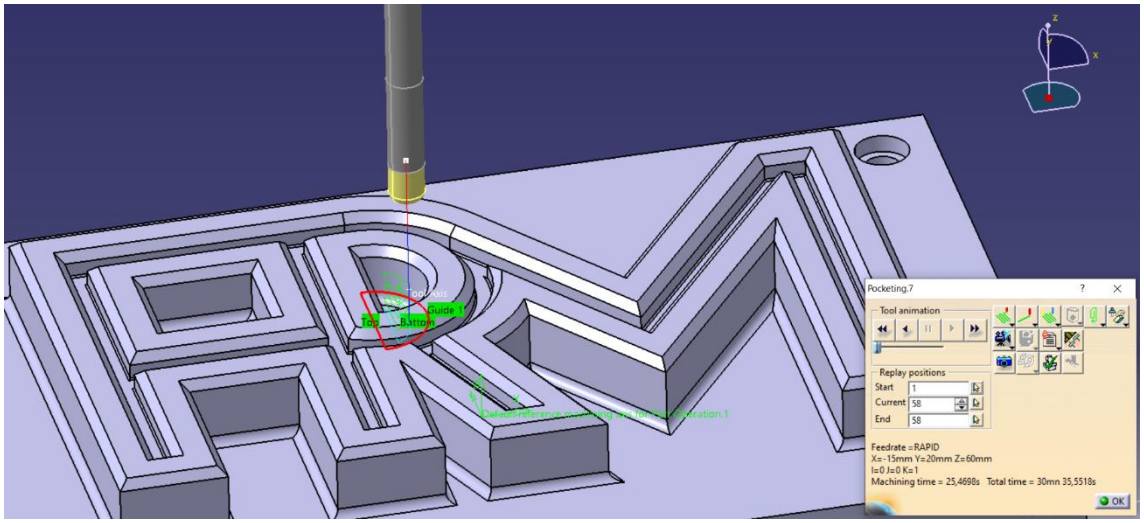


Contorneado con fresa tórica D10 Rc2.5.

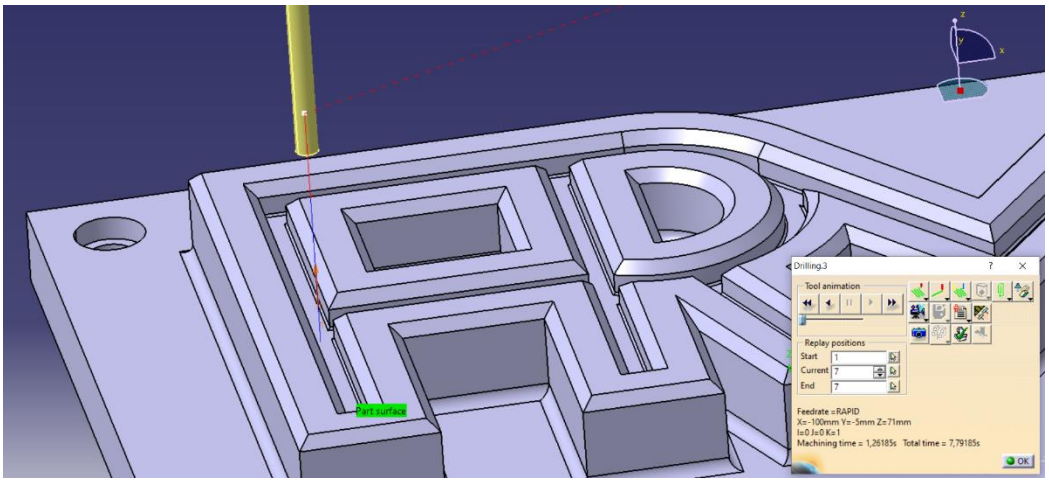


Desbaste del interior de las letras.

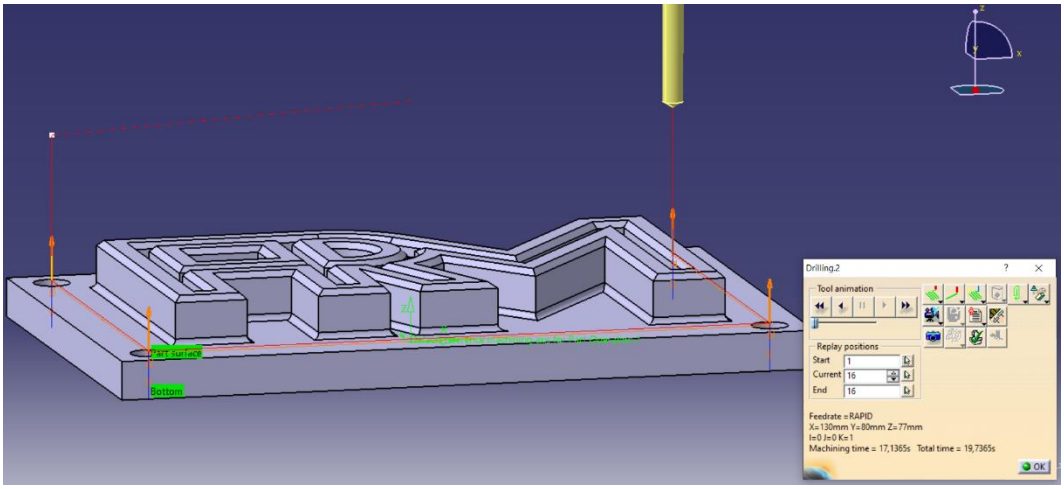




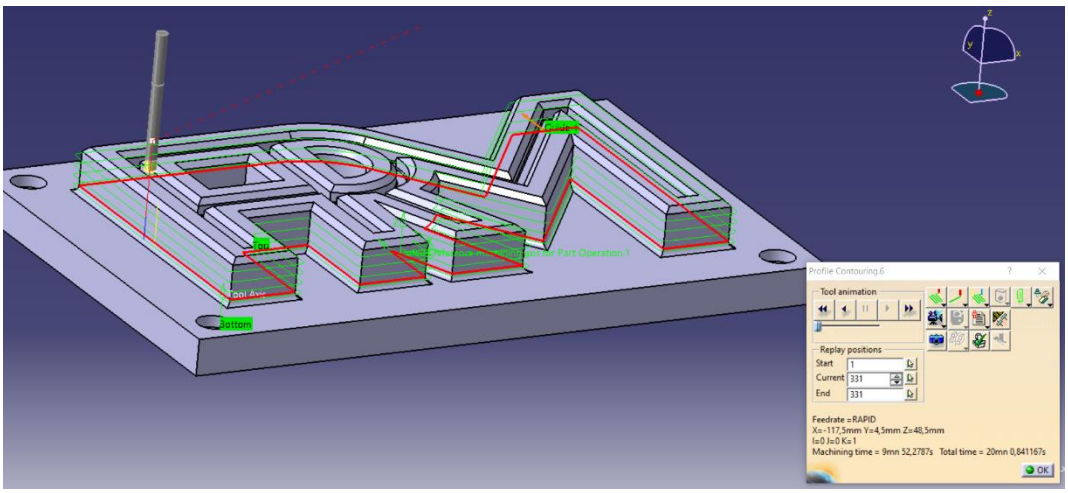
Taladro con broca D4.8 para posteriormente insertar la fresa D5 y hacer las ranuras.



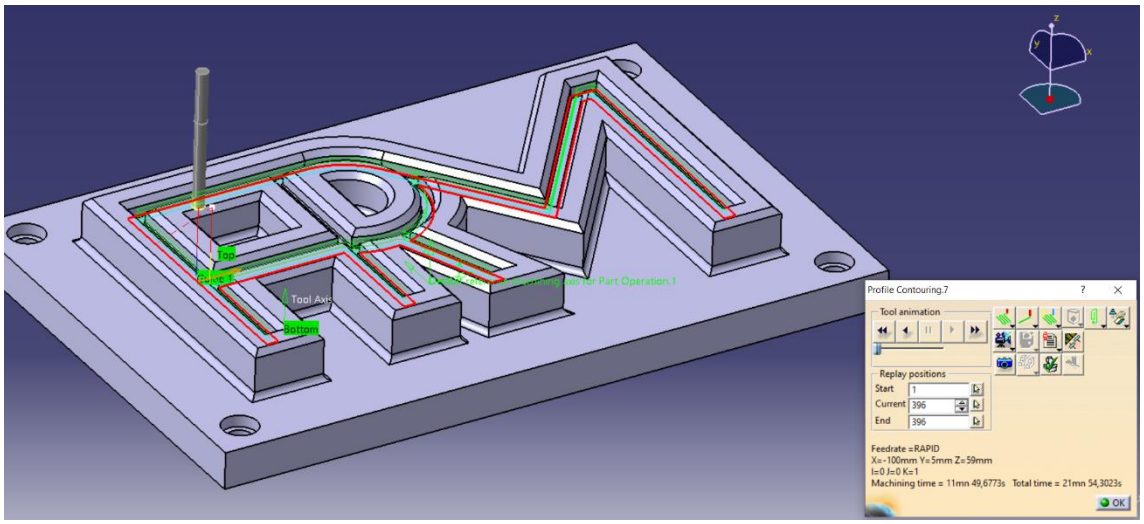
Taladros broca D8.5, hacemos un patrón para hacer los 4 agujeros en una sola operación.

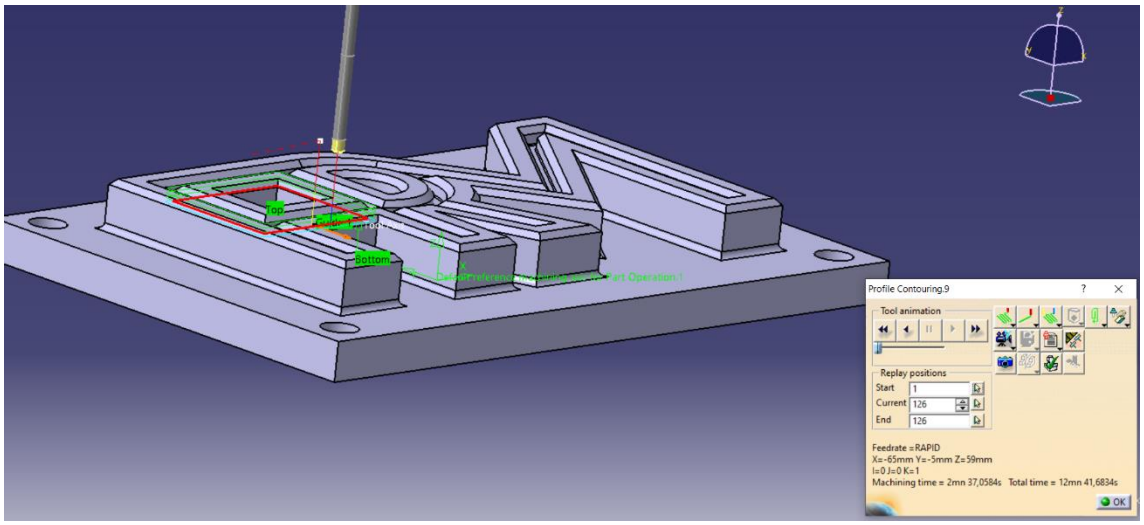


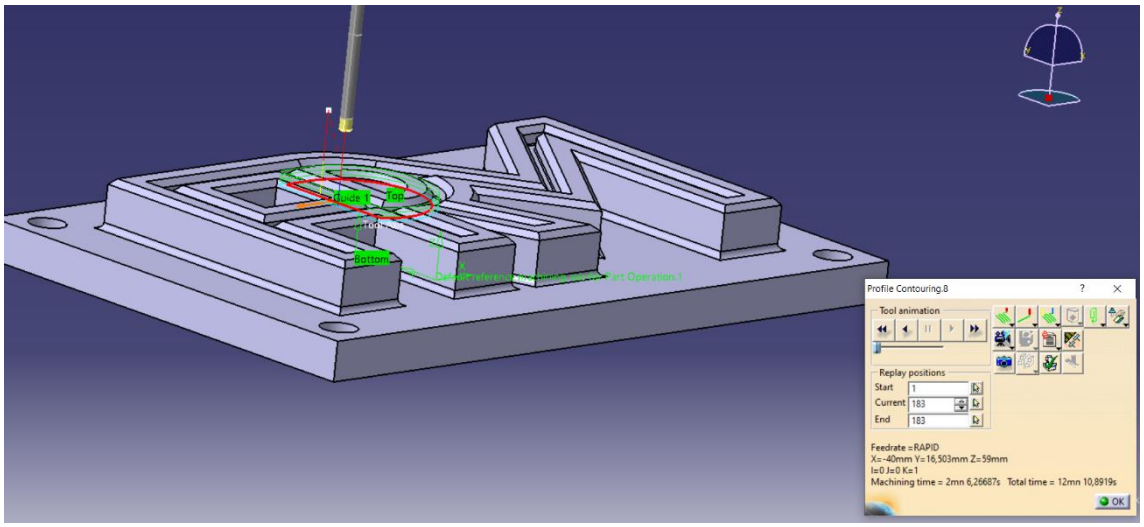
Contorneado con fresa D, usado únicamente para mejorar la precisión de nuestra pieza final, ya que por su geometría hay zonas donde las herramientas anteriores no podían acceder.



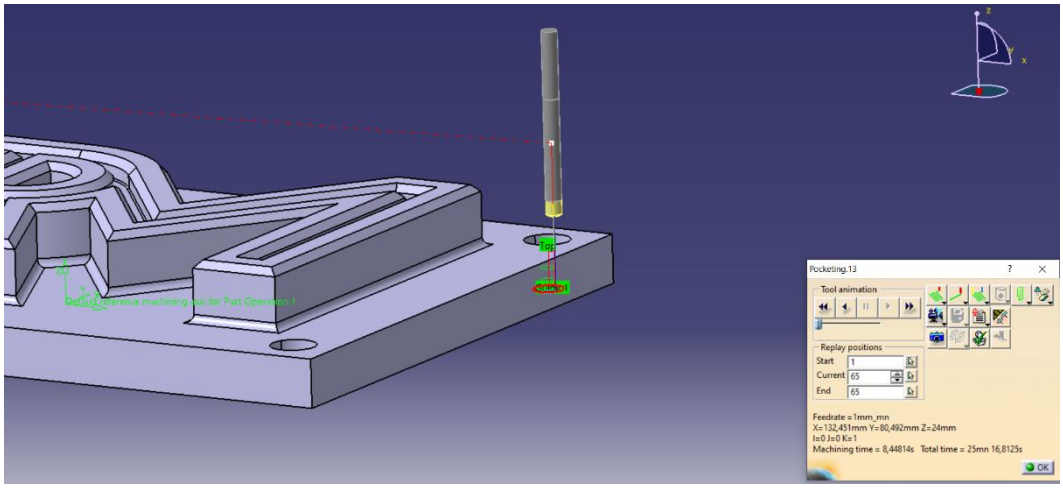
Contorneado para el interior de las letras, cuya profundidad es 7mm, usando la fresa tórica D5 Rc0.5, habiendo insertado la fresa por el agujero creado anteriormente por la broca D4.8.

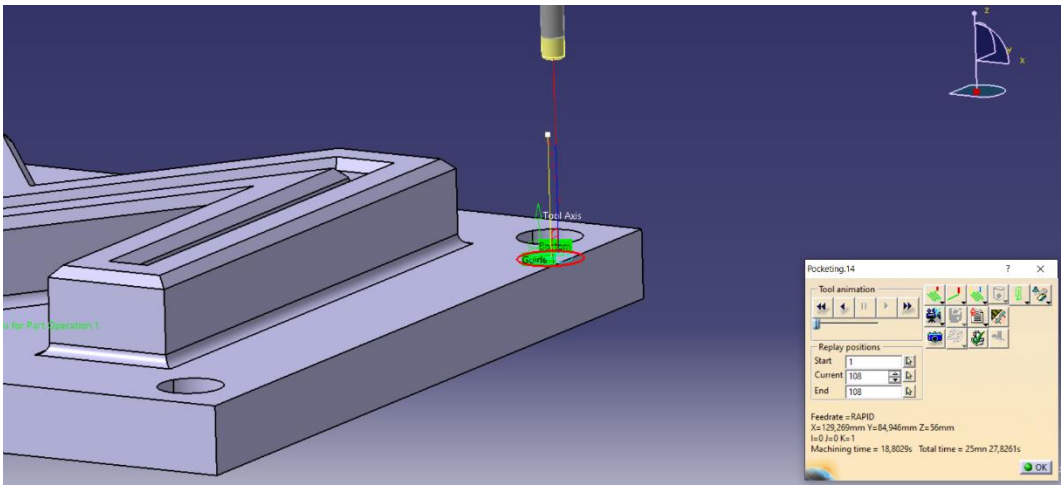




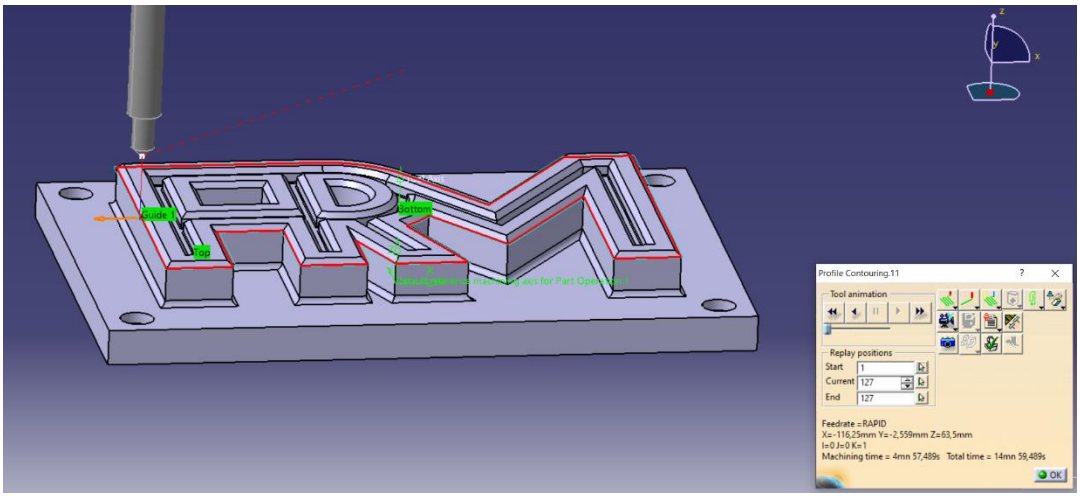


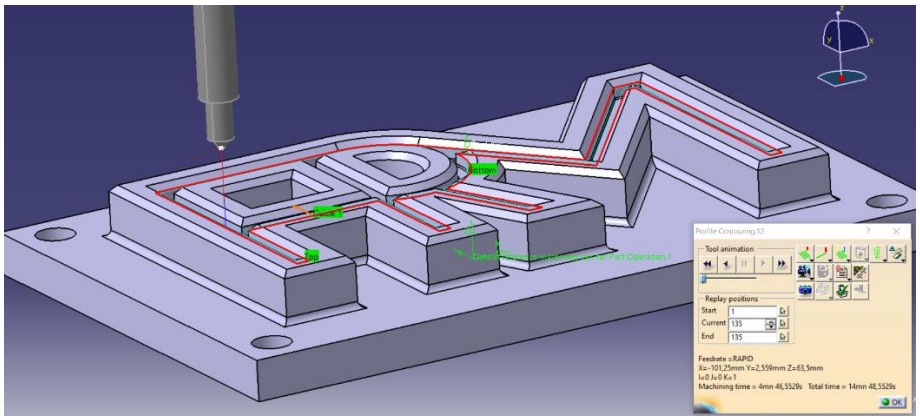
Apertura de los orificios de anclaje de la pieza con fresa tórica D5 Rc0.5 para que tuviesen el diámetro real, ya que no disponíamos directamente de taladros con un diámetro de 10mm. Para aprovechar el tiempo al máximo, la retirada de la fresa está a la distancia justa para el siguiente contorneado, dentro del mismo agujero pero con un diámetro de 15mm. Esta operación la repetimos cuatro veces en los cuatro agujeros de anclaje.

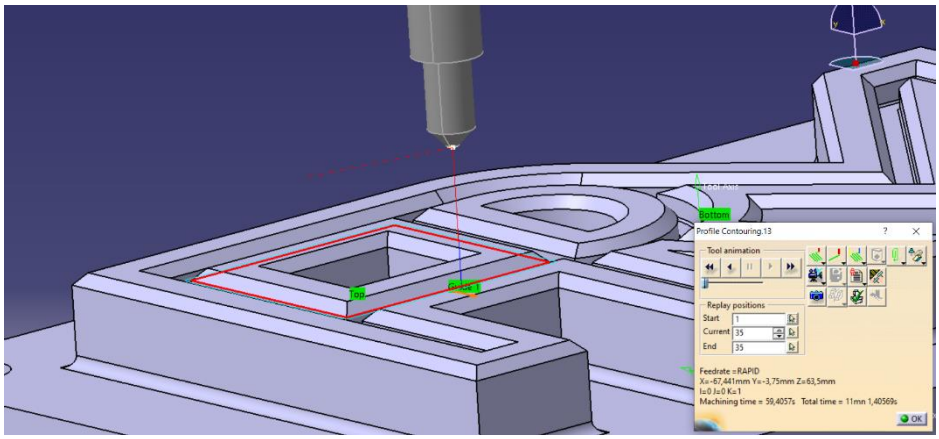


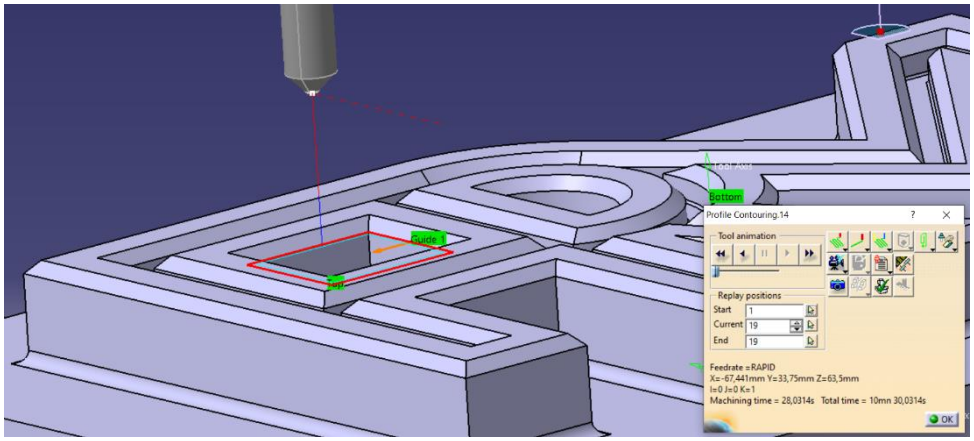


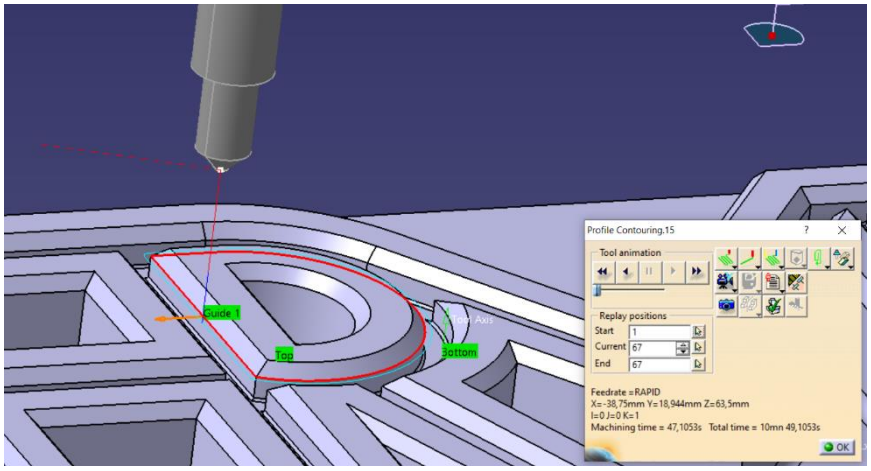
Contorneado con avellanador cónico de 90º



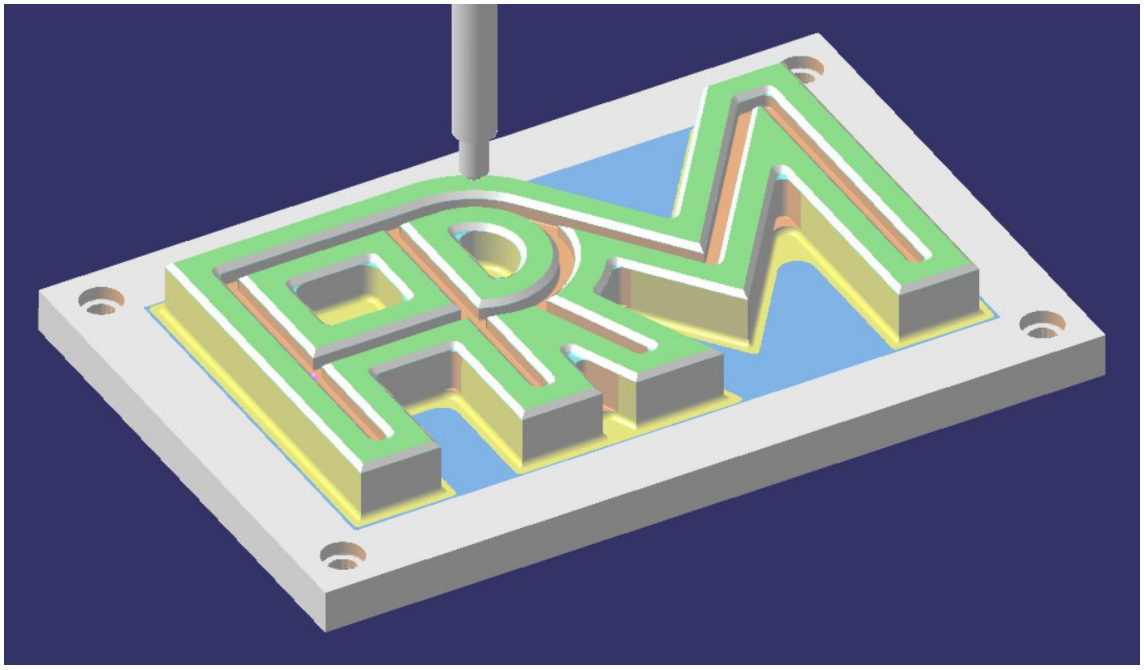

Profile Contouring.12
Tool animation
Replay positions
Start 1
Current 135
End 135
Feedrate =RAPID
X=-101,25mm Y=2,559mm Z=63,5mm
I=0 J=0 K=1
Machining time = 4mn 46,5529s Total time = 14mn 48,5529s
OK


Bottom
Profile Contouring.13
Tool animation
Replay positions
Start 1
Current 35
End 35
Feedrate =RAPID
X=-67,441mm Y=-3,75mm Z=63,5mm
I=0 J=0 K=1
Machining time = 59,4057s Total time = 11mn 1,40569s
OK


Bottom
Profile Contouring.14
Tool animation
Replay positions
Start 1
Current 19
End 19
Feedrate =RAPID
X=-67,441mm Y=33,75mm Z=63,5mm
I=0 J=0 K=1
Machining time = 28,0314s Total time = 10mn 30,0314s
OK


Profile Contouring.15
Tool animation
Replay positions
Start 1
Current 67
End 67
Feedrate =RAPID
X=-38,75mm Y=18,944mm Z=63,5mm
I=0 J=0 K=1
Machining time = 47,1053s Total time = 10mn 49,1053s
OK

5. Resultado final tras la mecanización de la misma



7. Tabla de procesos

Process Table for Manufacturing Program.1

View

View1

Number	Name	Type	Machining feedrate	Spindle speed	Spindle number	Resources name	Comment	Status	Locked	Computed	Packed	Video result	Specified time	Total time	Machining time	Machini
	Manufacturing Progr...	Manufacturing Program	-	-	-	-	-	-	-	-	-	No	5s	0s	0s	-
1	Tool Change.2	Tool Change	-	-	-	-/T1 Fresa de planear D80	-	-	-	-	-	No	5s	0s	0s	-
2	Facing.2	Facing	198,944mm_mn	795,775turn_mn Angular	-	-/T1 Fresa de planear D80	-	U...	No	Computed	No	No	5s	696,8s	434,294s	0,1mm
3	Tool Change.4	Tool Change	-	-	-	-/T5 Fresa D16	-	-	-	-	-	No	5s	0s	0s	-
4	Pocketing.1	Pocketing	5968,31mm_mn	3978,874turn_mn Angular	-	-/T5 Fresa D16	-	U...	No	Computed	No	No	5s	1960,163s	722,899s	0,1mm
5	Tool Change.5	Tool Change	-	-	-	-/T9 Fresa tórica D10 Rc2.5	-	-	-	-	-	No	5s	0s	0s	-
6	Profile Contouring.2	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/T9 Fresa tórica D10 Rc2.5	-	U...	No	Computed	No	No	5s	1241,416s	1318,91s	0,1mm
7	Pocketing.6	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T9 Fresa tórica D10 Rc2.5	-	U...	No	Computed	No	No	5s	1874,928s	59,9872s	0,1mm
8	Pocketing.7	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T9 Fresa tórica D10 Rc2.5	-	U...	No	Computed	No	No	5s	1835,552s	25,4698s	0,1mm
9	Tool Change.6	Tool Change	-	-	-	-/T12 Broca D4.8	-	-	-	-	-	No	5s	0s	0s	-
10	Drilling.3	Drilling	663,146mm_mn	3315,728turn_mn Angular	-	-/T12 Broca D4.8	-	U...	No	Computed	No	No	5s	7,792s	1,26185s	-
11	Tool Change.9	Tool Change	-	-	-	-/T11 Broca D8.5	-	-	-	-	-	No	5s	0s	0s	-
12	Drilling.2	Drilling	374,482mm_mn	1872,411turn_mn Angular	-	-/T11 Broca D8.5	-	U...	No	Computed	No	No	5s	19,737s	17,1365s	-
13	Tool Change.7	Tool Change	-	-	-	-/T3 Fresa tórica D5 Rc0.5	-	-	-	-	-	No	5s	0s	0s	-
14	Profile Contouring.6	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	875,088s	592,279s	0,1mm
15	Profile Contouring.7	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	910,273s	709,677s	0,1mm
16	Profile Contouring.9	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	672,268s	157,058s	0,1mm
17	Profile Contouring.8	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	659,006s	126,267s	0,1mm
18	Pocketing.13	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	1516,813s	8,44814s	0,1mm
19	Pocketing.14	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	1527,826s	18,8029s	0,1mm
20	Pocketing.12	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	1517,813s	8,44814s	0,1mm
21	Pocketing.16	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	1527,826s	18,8029s	0,1mm
22	Pocketing.20	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	1517,813s	8,44814s	0,1mm
23	Pocketing.19	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	1527,826s	18,8029s	0,1mm
24	Pocketing.11	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	1517,813s	8,44814s	0,1mm
25	Pocketing.18	Pocketing	450mm_mn	4500turn_mn Angular	-	-/T3 Fresa tórica D5 Rc0.5	-	U...	No	Computed	No	No	5s	1527,826s	18,8029s	0,1mm
26	Tool Change.8	Tool Change	-	-	-	-/Avellanador cónico 90°	-	-	-	-	-	No	5s	0s	0s	-
27	Profile Contouring.11	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/Avellanador cónico 90°	-	U...	No	Computed	No	No	5s	661,786s	297,489s	0,1mm
28	Profile Contouring.12	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/Avellanador cónico 90°	-	U...	No	Computed	No	No	5s	659,589s	286,553s	0,1mm
29	Profile Contouring.13	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/Avellanador cónico 90°	-	U...	No	Computed	No	No	5s	613,939s	59,4057s	0,1mm
30	Profile Contouring.14	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/Avellanador cónico 90°	-	U...	No	Computed	No	No	5s	607,633s	28,0314s	0,1mm
31	Profile Contouring.15	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/Avellanador cónico 90°	-	U...	No	Computed	No	No	5s	611,467s	47,1053s	0,1mm
32	Profile Contouring.16	Profile Contouring	450mm_mn	4500turn_mn Angular	-	-/Avellanador cónico 90°	-	U...	No	Computed	No	No	5s	605,915s	19,4816s	0,1mm

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